Tokenizer learning curve

Scholarly/code pretraining • 2026-10-16

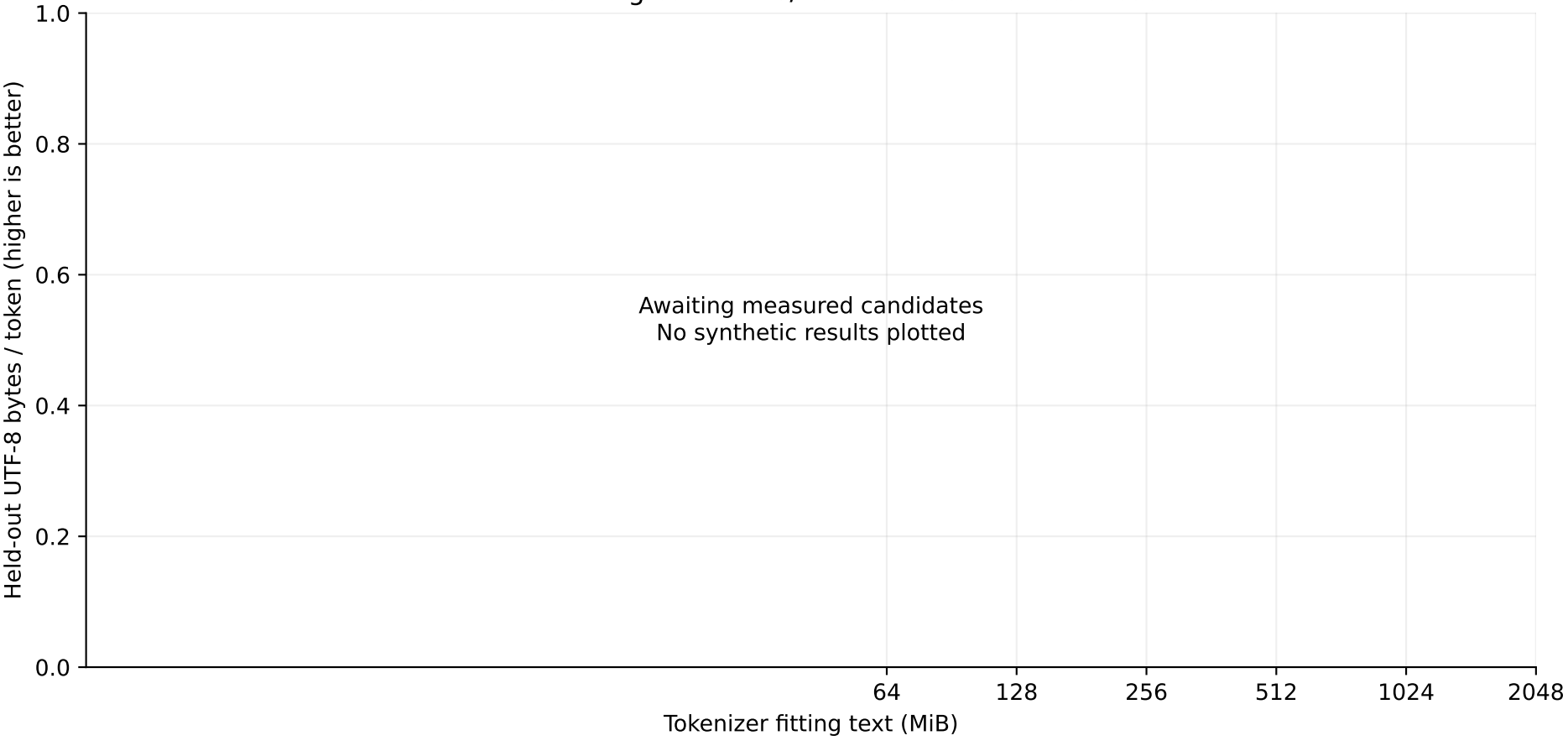
0 of 18 requested candidates have measured Modal results. The grid doubles fitting text from 64 MiB through 2 GiB and compares 16,384, 32,768 and 50,000 vocabulary entries. All fresh candidates use the same source-weighted, parent-held-out evaluation corpus. A higher bytes-per-token value means fewer tokens for the same text; it does not establish better language-model quality.

Design: six fitting sizes × three vocabulary sizes, nested samples from 12 pinned sources, and a shared 32 MiB held-out evaluation corpus.

Metrics: bytes per token by source and overall; round trips and unknown tokens; vocabulary utilization and fixed-term fragmentation; fitting time and peak memory.

Interpretation: compression is a proxy. Evaluate source regressions and compute tradeoffs before choosing a tokenizer. No downstream model-quality result is implied.

Limitations: one sampling seed; no global fuzzy near-deduplication guarantee; cross-source parent reconciliation remains incomplete. The complete methodology and numeric comparisons are in REPORT.md and measurements.json.


Tokenizer learning curve — 0/18 Modal candidates measured
Held-out UTF-8 bytes / token (higher is better)
1.0
0.8
0.6
0.4
0.2
0.0
Awaiting measured candidates
No synthetic results plotted
64
128
256
512
1024
2048
Tokenizer fitting text (MiB)